It is not required that your answers be submitted *on* this worksheet, but you may choose to do so; if you do, this pdf has been designed so that you can select "Assign Pages Sequentially" in Gradescope. If you choose to submit in another format, then: when submitting your questions, be sure that

each answer is submitted to the correct question

(you can change the order of pages within GradeScope) and that the page is

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your question will not be graded.

Your work may be submitted on your own paper, by writing on this worksheet printed out, or by writing via tablet, among possibly other options. If you are having problems submitting your work, please attend office hours. (There is no late penalty for submitting the assignment on the due date, but there will be no late submissions accepted for any reason.) Click on Gradescope's help page and read their instructions.

1. The pair of equations

$$\frac{dx}{dt} = -500x + xy$$
$$\frac{dy}{dt} = -600y + 3xy$$

models two coexisting species with populations $x(t)$ and $y(t)$. One equilibrium solution is the pair of constant functions $x(t) = 0$, $y(t) = 0$, representing both species being extinct. Find an equilibrium solution for this model where both species have positive population.

2. In an engineering project, your colleague comes up with a model $\frac{dy}{dx} = x^2 + y^4$. Based on experiments, your team has determined that the function you are looking for (a.k.a. the solution $y(x)$ to the model) cannot have any local maximum. Determine if the model has this property. Hint: use the First Derivative Test.

3. Find the volume of the solid obtained by rotating the region enclosed by $y = \dfrac{1}{x^2+4}$, $y = 0$, $x = 0$, and $x = 5$ around the y-axis.

   (a) Use the shell method.

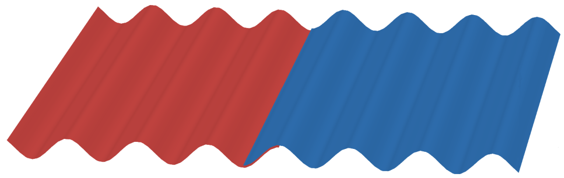
   (b) Use the disk/washer method.

4. Find the volume of the solid obtained by rotating the region enclosed by $y = \dfrac{1}{x^2+4}$, $y = 0$, $x = 0$, and $x = 5$ around the x-axis.

   (a) Use the disk/washer method.

   (b) Set up, but do not evaluate, the volume as an integral using the shell method.

5. A manufacturer wants to produce metal sheets for roofing by molding flat sheets into a sinusoidal $y = \sin(\pi x/9)$. If the final sheets are intended to be 22.5 inches wide, how wide should the initial flat sheets be? Use the midpoint rule with $n = 5$ subdivisions to approximate your integral to four decimal places. (See the illustration below of two interlocked metal sheets.)



6. Cables hanging between two poles (for example, elevated power lines) follow a specific curve. In the coordinate system where the origin is located on the ground halfway between the poles (located at $x = \pm B$), the curve has equation

$$y = C + A\cosh\left(\frac{x}{A}\right) \,.$$

(a) Find an integral expression for the length of the cable, and evaluate that integral. (Your final answer may be in terms of A, B, and/or C.)

(b) Suppose that the two poles are 80 feet apart, the cable is 81 feet long, and that the lowest point of the cable must be 25 feet above the ground (to avoid tall trucks, etc.). How tall must the poles be? Give your answer to two decimal places. (You will need to numerically estimate some values in this calculation. You can use e.g. Desmos.)